Supplementary Verification of American-Value Convergence

Application of Kushner–Dupuis, Chapter 10, Theorem 6.2

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Abstract

This note supplies the detailed justification for Step 3 of Theorem 4 of *Binomial Trees for Optimal Stopping Problems under Drift Switching and Partial Information*. We record the setup, assumptions, and conclusion of Kushner and Dupuis, Chapter 10, Theorem 6.2, including the standing conditions on the approximating chains. We then verify each hypothesis for the bounded, localized stopping problem. Particular attention is given to exit-time continuity, admissible filtrations, the distinction between deterministic and exponential holding times, and exact identification of the discrete values. The final section covers the randomized filter-grid approximation used in Step 4.

1 The Kushner–Dupuis theorem and its standing setup

All book references and page numbers below refer to the second edition of Kushner and Dupuis [1]. The statement below is a restatement, with inherited hypotheses made explicit, rather than a new convergence theorem or a verbatim quotation. We retain the book's notation $x(t), w(t), \xi_n^h, p^h, \Delta t^h, k, g, V, V^h$. The letter y denotes a generic initial state, to distinguish it from the initial asset price x_0 used in Section 2.

1.1 Continuous process, costs, and obligatory exit

Section 10.6, equation (6.1), pp. 296–297, uses the uncontrolled jump-diffusion

$$\begin{aligned} x(t) = y &+ \int_0^t b(x(s))\,\mathrm{d}s + \int_0^t \sigma_{\mathrm{KD}}(x(s))\,\mathrm{d}w(s) \\ &+ \int_0^t \int_\Gamma q(x(s-),p)\,N(\mathrm{d}s,\,\mathrm{d}p). \end{aligned} \tag{1.1}$$

Here $x(t) \in \mathbb{R}^d$, w is a standard Brownian motion, and N is a Poisson random measure with finite intensity $\lambda\,\mathrm{d}s\,\mathsf{P}(\mathrm{d}p)$; the mark distribution P has compact support Γ. These noises are standard relative to the admissible filtration, so their future noise is nonanticipating relative to its present information. The jump model is described in Sections 1.5 and 5.6.1, in particular pp. 127–128. The notation σ_{KD} distinguishes the book's dispersion matrix from the scalar asset volatility σ below.

Let $C \subset \mathbb{R}^d$ be the stopping domain; the book also calls this set G. Write C° for its interior and define

$$\tau_C = \inf\{t \geq 0 : x(t) \notin C^\circ\}, \qquad \tau_C' = \inf\{t \geq 0 : x(t) \notin C\}. \tag{1.2}$$

The two times are different definitions: the first includes contact with the boundary, whereas the second uses the exterior of the closed set. They will have the same infimum almost surely in our application. For an admissible stopping time $\rho \leq \tau_C$, the book's costs are

$$W(y,\rho) = \mathbb{E}_y\left[\int_0^\rho k(x(s))\,\mathrm{d}s + g(x(\rho))\right], \qquad V(y) = \inf_{\rho \leq \tau_C} W(y,\rho). \tag{1.3}$$

The infimum has the weak formulation used in the book: the probability space and filtration may vary, provided the noises remain standard, the solution is adapted, and ρ is a stopping time. Our application has no jumps, no running cost, and stopping times bounded by T.

1.2 Explicit assumptions inherited by Theorem 6.2

The following are the book's conditions, with control variables suppressed as in Section 10.6.

(A1.1) b and σ_{KD} are bounded and continuous.

(A1.2) q is bounded and measurable, and $q(\cdot, p)$ is continuous for every mark p.

(A1.3) The cost functions k, c, g are bounded and continuous. The function c occurs elsewhere in the book's cost formulations; it does not enter (1.3) and can be set to zero here.

(A2.1) C is compact and equals the closure of C°.

(A6.1) For every initial state in C°, the law of (w, N, ρ) determines the law of (x, w, N, ρ) for any admissible solution and stopping time. In addition, the first-exit functional

$$f(\phi) = \inf\{t \geq 0 : \phi(t) \notin C^\circ\}$$

on Skorokhod space $D([0,\infty); \mathbb{R}^d)$ is almost surely continuous under the solution law for each initial state of interest, or the book's alternative randomized exit procedure is used.

Conditions (A1.1)–(A1.3) are on p. 269, (A2.1) on p. 279, and (A6.1) on pp. 296–297. We use the exit-continuity alternative, not the randomized exit procedure. The remark after (A6.1), p. 297, states that its uniqueness requirement follows from a uniform Lipschitz condition on the coefficients. This requirement concerns the joint law with the noise and stopping time; uniqueness of the state's marginal law alone is not its definition.

1.3 The approximating chain and local consistency

For each approximation parameter h, let ξ_n^h be a Markov chain on a discrete state space S^h, with transition probabilities $p^h(y, y')$ and interpolation intervals $\Delta t^h(y)$ such that

$$\inf_{y \in S^h} \Delta t^h(y) > 0 \quad \text{for each } h, \qquad \sup_{y \in S^h} \Delta t^h(y) \longrightarrow 0. \tag{1.4}$$

The states in $S^h \cap C$ are finite in number for each h. For a diffusion, put $a = \sigma_{\mathrm{KD}} \sigma_{\mathrm{KD}}^\top$. Equation (4.1.3), p. 71, requires

$$\mathbb{E}_y[\Delta \xi^h] = b(y)\Delta t^h(y) + o(\Delta t^h(y)), \tag{1.5}$$

$$\mathrm{Cov}_y(\Delta \xi^h) = a(y)\Delta t^h(y) + o(\Delta t^h(y)), \tag{1.6}$$

$$\sup_{n,\omega} |\xi_{n+1}^h - \xi_n^h| \longrightarrow 0. \tag{1.7}$$

The conditional moments are given the current state, equivalently the chain history, and the errors are uniform over the relevant states on compact sets. These are covariance conditions, not merely raw-second-moment conditions.

For completeness, in the jump-diffusion version the standing construction is Section 5.6.2, equation (5.6.6), p. 129. Its transition probabilities have the form

$$\begin{aligned} p^h(y, y') = {} & [1 - \lambda \Delta t^h(y) - \delta^h(y)] p_D^h(y, y') \\ & + [\lambda \Delta t^h(y) + \delta^h(y)] \mathsf{P}\{p : q_h(y, p) = y' - y\}, \end{aligned} \tag{1.8}$$

where p_D^h satisfies the diffusion conditions (1.5)–(1.7), $\delta^h = o(\Delta t^h)$, and q_h is measurable, uniformly bounded, and converges to q uniformly in y on compact sets for each mark. Also $y + q_h(y,p) \in S^h$, and the mixture weights are probabilities. In (1.7), only the diffusion branch is intended when (1.8) is used: genuine Poisson jumps need not vanish. Our application sets $\lambda = 0$, $\delta^h = 0$, and has no jump branch, so (1.5)–(1.7) apply to the whole chain.

Local consistency is not repeated in the short statement of Theorem 6.2. It is part of the approximation setup recalled at the beginning of Section 10.4, p. 286, and explicitly assumed in Chapter 10, Theorem 4.1, p. 290, used in the proof of Theorem 6.2. In particular, these conditions allow nonzero conditional moment errors; the chain need not be exact sampling of a given diffusion.

1.4 Discrete costs and the convergence statement

Let

$$\nu_C^h = \inf\{n \geq 0 : \xi_n^h \notin C^\circ\}.$$

For stopping indices $J \leq \nu_C^h$ in the natural chain filtration, the discrete cost defined on p. 297 is

$$W^h(y,J) = \mathbb{E}_y\left[\sum_{n=0}^{J-1} k(\xi_n^h)\Delta t^h(\xi_n^h) + g(\xi_J^h)\right], \qquad V^h(y) = \inf_{J \leq \nu_C^h} W^h(y,J). \tag{1.9}$$

The terminal cost is evaluated at the actual exit state, including overshoot. On p. 298 the Bellman equation is

$$V^h(y) = \min\left\{g(y),\, k(y)\Delta t^h(y) + \sum_{y'} p^h(y,y')V^h(y')\right\}, \quad y \in S^h \cap C^\circ, \tag{1.10}$$

with $V^h = g$ outside C°.

Theorem 1.1 (Kushner–Dupuis, Chapter 10, Theorem 6.2, p. 298)**.** *Under the standing approximation setup above and conditions (A1.1)–(A1.3), (A2.1), and (A6.1), assume either*

$$\inf_{y \in C} k(y) > 0, \tag{1.11}$$

as in Theorem 6.1, or replace (1.11) *by*

$$\sup_{y \in C} \mathbb{E}_y[\tau_C'] < \infty. \tag{1.12}$$

Then, for each fixed initial state $y \in C^\circ$ represented in the approximating chains,

$$V^h(y) \longrightarrow V(y) \qquad (h \downarrow 0).$$

This makes explicit the hypotheses inherited from Theorem 6.1 (p. 297) and the standing chain construction. Theorem 6.2 expressly allows (1.12) to replace strict positivity of k. Thus setting $k = 0$ is legitimate when (1.12) is verified. We do not apply the positive-running-cost assertion of Theorem 6.1 to a zero-running-cost problem.

1.5 The two time interpolations

Section 4.2, pp. 72–74, uses prescribed transition times

$$t_n^h = \sum_{j=0}^{n-1} \Delta t^h(\xi_j^h), \qquad \xi^h(t) = \xi_n^h \quad (t_n^h \leq t < t_{n+1}^h).$$

The Chapter 10 proofs instead use the auxiliary continuous-time Markov chain of Section 4.3, pp. 75–77:

$$s_n^h = \sum_{j=0}^{n-1} \zeta_{j+1} \Delta t^h(\xi_j^h), \qquad \psi^h(t) = \xi_n^h \quad (s_n^h \le t < s_{n+1}^h),$$

where the ζ_j are independent unit-mean exponential random variables, independent of the embedded chain. These interpolations have the same sequence of states but different holding times. The book identifies their optimal stopping values by their common Bellman equation (1.10), pp. 297–298. Its convergence conclusion therefore applies to the discrete value (1.9), although its proof uses ψ^h. In our application all discounting is encoded in the calendar-time state coordinate, as explained below; there is no additional discount factor based on the auxiliary clock.

2 The stopping problem in the paper

We make explicit the relevant definitions from the main paper [3]. Fix $T > 0$, scalar volatility $\sigma > 0$, drifts μ_0, μ_1, and nonnegative switching intensities λ_0, λ_1. Write

$$\widehat{\mu}(p) = \mu_0(1-p) + \mu_1 p, \quad \Lambda(p) = \lambda_0(1-p) - \lambda_1 p, \quad \varrho = \frac{\mu_1 - \mu_0}{\sigma}.$$

The paper denotes ϱ by ρ; here ρ is reserved for the book's stopping time. The posterior-filtered log-price and probability solve

$$\mathrm{d}Z_t = (\widehat{\mu}(\Pi_t) - \sigma^2/2)\,\mathrm{d}t + \sigma\,\mathrm{d}\widehat{W}_t, \tag{2.1}$$

$$\mathrm{d}\Pi_t = \Lambda(\Pi_t)\,\mathrm{d}t + \varrho\Pi_t(1-\Pi_t)\,\mathrm{d}\widehat{W}_t, \tag{2.2}$$

with $Z_0 = z_0 = \log x_0$, $x_0 > 0$, and $\Pi_0 = \pi_0 \in [0,1]$. Here $\widehat{W}$ is the innovation Brownian motion and $X_t = \mathrm{e}^{Z_t}$. The paper's Lemma 1 gives a unique global strong solution with $\Pi_t \in [0,1]$.

Let $G : [0,T] \times (0,\infty) \to [0,\infty)$ be continuous, with

$$G(t,x) \le C_G(1 + x^\gamma), \qquad \gamma \ge 1.$$

The exponent is denoted q in the paper; we use γ here to avoid confusion with the book's jump coefficient q. For fixed $M > 0$, $R > |z_0|$, and $r \ge 0$, set

$$G^M(t,x) = G(t, x \wedge M), \qquad K_M = C_G(1 + M^\gamma), \tag{2.3}$$

$$\vartheta_R = T \wedge \inf\{t \ge 0 : |Z_t| \ge R\}. \tag{2.4}$$

The bounded, localized continuous value in Step 3 is

$$V_R^M(x_0,\pi_0) = \sup_{\rho \in \mathcal{T}^X,\ 0 \le \rho \le \vartheta_R} \mathbb{E}_{x_0,\pi_0}\left[\mathrm{e}^{-r\rho} G^M(\rho, \mathrm{e}^{Z_\rho})\right], \tag{2.5}$$

where $\mathcal{T}^X$ denotes observation-filtration stopping times.

For the discrete model take $h = T/N$ sufficiently small that the transition and emission probabilities below are valid:

$$u_h = \mathrm{e}^{\sigma\sqrt{h}}, \qquad d_h = \mathrm{e}^{-\sigma\sqrt{h}}, \tag{2.6}$$

$$p_{u,i}^h = \frac{\mathrm{e}^{\mu_i h} - d_h}{u_h - d_h}, \qquad p_{d,i}^h = 1 - p_{u,i}^h \qquad (i = 0,1), \tag{2.7}$$

$$\bar{p}_h = p + h\Lambda(p), \qquad p_\ell^h(p) = p_{\ell,0}^h(1-\bar{p}_h) + p_{\ell,1}^h \bar{p}_h, \tag{2.8}$$

$$\Phi_\ell^h(p) = \frac{p_{\ell,1}^h \bar{p}_h}{p_\ell^h(p)}, \qquad \ell \in \{u,d\}. \tag{2.9}$$

Starting from $(Z_0, \Pi_0) = (z_0, \pi_0)$, select branch ℓ with probability $p^h_\ell(\Pi_n)$ and set

$$(Z_{n+1}, \Pi_{n+1}) = (Z_n + \epsilon_\ell \sigma\sqrt{h},\ \Phi^h_\ell(\Pi_n)), \qquad \epsilon_u = 1, \quad \epsilon_d = -1. \tag{2.10}$$

The superscripts h make explicit the dependence suppressed in the paper's p_ℓ and Φ_ℓ. The chain is continued beyond $n = N$ with the same transition rule when an unrestricted chain is needed for the approximation setup. Define

$$\nu^{(N)}_R = N \wedge \inf\{n \geq 0 : |Z_n| \geq R\}.$$

With J ranging over return-filtration stopping indices,

$$v^{(N),M}_R(x_0, \pi_0) = \sup_{J \leq \nu^{(N)}_R} \mathbb{E}_{x_0,\pi_0}\left[\mathrm{e}^{-rJh} G^M(Jh, \mathrm{e}^{Z_J})\right]. \tag{2.11}$$

The notation Π_n refers to the discrete filter in (2.10), whereas Π_t in (2.2) refers to the continuous filter. Discrete indices and continuous times distinguish these uses.

3 Identification with the book's setup

Use the book's state, Brownian motion, chain, and domain as follows:

$$x(t) = (t, Z_t, \Pi_t), \qquad \xi^h_n = (nh, Z_n, \Pi_n), \qquad C = [-1, T] \times [-R, R] \times [-1, 2]. \tag{3.1}$$

The initial state is $y_0 = (0, z_0, \pi_0) \in C^\circ$. The first coordinate turns the finite-horizon problem into an autonomous stopping problem. The dimension is three because time, log-price, and posterior probability are its three state coordinates; the theorem does not prescribe this dimension.

For globally defined coefficients put $\kappa(p) = (0 \vee p) \wedge 1$:

$$\begin{aligned} b(s, z, p) &= \begin{pmatrix} 1 \\ \widehat{\mu}(\kappa(p)) - \sigma^2/2 \\ \Lambda(\kappa(p)) \end{pmatrix}, \\ \sigma_{\mathrm{KD}}(s, z, p) &= \begin{pmatrix} 0 & 0 & 0 \\ \sigma & 0 & 0 \\ \varrho\kappa(p)(1 - \kappa(p)) & 0 & 0 \end{pmatrix}. \end{aligned} \tag{3.2}$$

Take $w = (\widehat{W}, B_2, B_3)$, adjoining two independent Brownian motions if the book's square-matrix convention is used. Only the first component drives the state. Equivalently one may use the single nonzero column and scalar Brownian motion $\widehat{W}$. Section 5 explains why adjoining independent, nonanticipating information does not alter the stopping value.

Set the book's Poisson jump intensity to zero and its jump coefficient to zero. Although the hidden regime can jump, the filtered state (2.1)–(2.2) is continuous: the hidden regime itself is not a state coordinate here. Set $\Delta t^h(y) = h$ and let p^h be the kernel of (3.1) induced by (2.10); explicitly,

$$p^h\left((s, z, p),\ (s + h, z + \epsilon_\ell \sigma\sqrt{h}, \Phi^h_\ell(p))\right) = p^h_\ell(p), \qquad \ell \in \{u, d\}. \tag{3.3}$$

This kernel is needed on the reachable chain states, not at arbitrary points of $\mathbb{R}^3$ outside the posterior strip.

Let $\chi(s) = (0 \vee s) \wedge T$ and define the book's costs by

$$k = c = 0, \qquad g(s, z, p) = -\mathrm{e}^{-r\chi(s)} G^M(\chi(s), \mathrm{e}^z). \tag{3.4}$$

In particular $|g| \le K_M$ on all of $\mathbb{R}^3$. Along the chain, for $n \le \nu_R^{(N)}$,

$$g(\xi_n^h) = -\mathrm{e}^{-rnh} G^M(nh, X_n). \tag{3.5}$$

Thus the discrete cost agrees exactly with the negative of the reward in (2.11), including the actual spatial overshoot. Discounting uses the coordinate nh, not the random elapsed time of the auxiliary interpolation.

The time coordinate increases with speed one in the diffusion and by h per chain transition. The posterior stays in $[0, 1]$. Consequently, from y_0,

$$\tau_C = \vartheta_R, \qquad \nu_C^h = \nu_R^{(N)}. \tag{3.6}$$

The process and kernel remain unrestricted: obligatory exit restricts admissible stopping rules. We do not multiply the coefficients by a discontinuous indicator of C or apply the theorem to a new SDE with absorbing coefficients.

4 Verification of all hypotheses

4.1 Conditions (A1.1)–(A1.3) and (A2.1)

The clipping function κ is globally Lipschitz and bounded. The functions $\widehat{\mu}$ and Λ are affine, and $\kappa(p)(1 - \kappa(p))$ is bounded and globally Lipschitz. Thus (3.2) is bounded and globally Lipschitz, in particular continuous, verifying (A1.1). Condition (A1.2) holds trivially since the Poisson jump term is absent. The costs (3.4) are bounded and continuous, verifying (A1.3). Only continuity and the polynomial bound on G are used; no reward derivatives are assumed. Finally,

$$C^\circ = (-1, T) \times (-R, R) \times (-1, 2), \qquad \overline{C^\circ} = C,$$

and C is compact, so (A2.1) holds.

4.2 Interpolation intervals and finiteness of the state set

Since $\Delta t^h = h = T/N$, condition (1.4) is immediate. For fixed N the chain has at most 2^n reachable states at step n, one for each possible sequence of n returns. The calendar-time coordinate is nh, and at step N it reaches T, forcing exit from C°. Therefore the number of reachable states in C is at most

$$\sum_{n=0}^{N} 2^n = 2^{N+1} - 1 < \infty.$$

The chain can still be continued for all $n \ge 0$; its calendar-time coordinate then exceeds T, so subsequent states are outside C.

4.3 Local consistency and its uniform error bounds

The computational input is Lemma 2, the local-consistency lemma in the main paper. We restate precisely what is used, without repeating its Taylor-remainder calculation. Uniformly over $p \in [0, 1]$, for some constant K independent of h and p,

$$\mathbb{E}_p[\Delta Z] = (\widehat{\mu}(p) - \sigma^2/2)h + O(h^2), \tag{4.1}$$

$$(\Delta Z)^2 = \sigma^2 h, \qquad \mathbb{E}_p[\Delta \Pi] = \Lambda(p)h, \tag{4.2}$$

$$\mathbb{E}_p[(\Delta \Pi)^2] = \varrho^2 p^2 (1 - p)^2 h + O(h^{3/2}), \tag{4.3}$$

$$\mathbb{E}_p[\Delta Z\, \Delta \Pi] = \sigma \varrho p(1 - p)h + O(h^{3/2}), \tag{4.4}$$

$$|\Delta \Pi| \le K\sqrt{h}. \tag{4.5}$$

Each $O(h^\alpha)$ means an error whose absolute value is at most Kh^α for all sufficiently small h and every $p \in [0,1]$. The estimates are also uniform in z and the calendar-time coordinate because the one-step increment law does not depend on those coordinates.

Write $m_h = \mathbb{E}_y[\Delta\xi^h]$. Equations (4.1)–(4.4), with the deterministic time increment h, give

$$m_h = h\,b(y) + O(h^2), \qquad |m_h| \le K_1 h.$$

The time increment has zero conditional variance and zero conditional covariance with every other coordinate. For the remaining two coordinates, raw second moments are converted to covariances by

$$\mathrm{Cov}_y(\Delta\xi^h) = \mathbb{E}_y[\Delta\xi^h(\Delta\xi^h)^\top] - m_h m_h^\top.$$

Since $m_h m_h^\top = O(h^2)$, the resulting conditional covariance is

$$\mathrm{Cov}_y(\Delta\xi^h) = h\,\sigma_{\mathrm{KD}}(y)\sigma_{\mathrm{KD}}(y)^\top + O(h^{3/2}).$$

For matrices these errors are bounded in maximum-entry norm. After division by h, both mean and covariance errors vanish uniformly. Moreover, for $h \le 1$,

$$|\Delta\xi^h| \le h + |\Delta Z| + |\Delta\Pi| \le h + (\sigma + K)\sqrt{h} \longrightarrow 0$$

uniformly. This verifies every part of the diffusion local-consistency requirement (1.5)–(1.7). No exact pathwise sampling or common probability space is assumed.

4.4 The strengthened uniqueness part of (A6.1)

The bounded globally Lipschitz coefficients (3.2) have a unique global strong solution from every initial state in $\mathbb{R}^3$; see [2, Chapter 5, Theorems 2.5 and 2.9]. The solution is a measurable nonanticipative function of its initial state and Brownian path. Therefore, for every adjoined admissible stopping time ρ, the law of (w,ρ) determines that of (x,w,ρ). This is precisely the no-jump uniqueness requirement of (A6.1), including initial states in C° outside the original posterior strip. It is also the Lipschitz sufficient condition stated after (A6.1) in [1, p. 297].

From y_0 the extended solution agrees with (2.1)–(2.2), by pathwise uniqueness and the strip invariance established in the paper's Lemma 1. Global Lipschitz continuity is used for uniqueness here; it does not by itself prove continuity of the exit functional.

4.5 Almost-sure continuity of the exit functional

Starting from y_0, the time coordinate cannot reach its lower bound -1, and $\Pi_t \in [0,1]$ cannot reach the artificial bounds -1 or 2. Thus obligatory exit is at the terminal time T or when Z first reaches $-R$ or R. The time coordinate immediately exceeds T after that time.

For the spatial boundary put $\eta_R = \inf\{t \ge 0 : |Z_t| \ge R\}$. On $\{\eta_R < T\}$,

$$Z_{\eta_R+u} - Z_{\eta_R} = \sigma(\widehat{W}_{\eta_R+u} - \widehat{W}_{\eta_R}) + D_u, \qquad |D_u| \le Bu$$

for a deterministic bound B on the drift. By the strong Markov property of Brownian motion, the post-η_R Brownian increments have the usual small-time oscillations. Their excursions of both signs dominate the $O(u)$ drift term at arbitrarily small positive times. If $Z_{\eta_R} = R$, there are therefore times arbitrarily close to η_R with $Z > R$; if $Z_{\eta_R} = -R$, there are such times with $Z < -R$. The same conclusion at a simultaneous spatial and terminal-time exit follows already from the time coordinate. This is the directional non-tangency mechanism discussed in [1, Section 10.2, pp. 278–280]; full nondegeneracy of the three-dimensional covariance is not needed.

For clarity, the deterministic continuity argument is as follows. Fix a continuous path x with the properties just described, started in C°, and put $\tau = f(x) < \infty$. Suppose $\phi_j \to x$ in the Skorokhod J_1 topology. At a continuous limit this implies uniform convergence on compact time intervals. For $0 < \varepsilon < \tau$, the compact path segment $x([0, \tau - \varepsilon])$ has positive distance from the complement of C°, so $f(\phi_j) > \tau - \varepsilon$ eventually. There is also $t \in (\tau, \tau + \varepsilon)$ with $x(t) \notin C$. Since C is closed, $x(t)$ has positive distance from C, and eventually $\phi_j(t) \notin C$. Hence $f(\phi_j) \le t < \tau + \varepsilon$ eventually. It follows that $f(\phi_j) \to f(x)$.

These properties hold almost surely for the limiting diffusion from y_0, proving the continuity part of (A6.1). In particular, $\tau'_C = \tau_C = \vartheta_R$ as infima of exit times from the closed and open domains, respectively. Continuation past T is useful for this path-space argument, even though every admissible stopping rule must stop by T.

4.6 The exit-moment alternative, including the auxiliary clock

For any initial state $(s, z, p) \in C$, the first coordinate is $s + t$. Consequently

$$\tau'_C \le T - s \le T + 1, \qquad \sup_{y \in C} \mathbb{E}_y[\tau'_C] \le T + 1.$$

This verifies (1.12), independently of the spatial and filter dynamics. We therefore use the second alternative in Theorem 1.1; the fact that $k = 0$ causes no difficulty.

There is also a direct moment bound for the auxiliary exponential-clock exit time, denoted here by τ^h_{aux}. Starting from y_0, exit occurs in at most N transitions, so, with the holding times of Section 1.5,

$$\tau^h_{\mathrm{aux}} \le h \sum_{j=1}^{N} \zeta_j, \qquad \mathbb{E}\left[h \sum_{j=1}^{N} \zeta_j\right] = T, \qquad \mathbb{E}\left[\left(h \sum_{j=1}^{N} \zeta_j\right)^2\right] = T^2 + Th.$$

Thus the auxiliary exit times, and stopping times bounded by them, are uniformly integrable. They need not be bounded by T: it is the calendar-time state coordinate nh, not the auxiliary elapsed time, that reaches T after N transitions. Uniform bounds over all initial $s \in [-1, T]$ follow in the same way by replacing N with $\lceil (T + 1)/h \rceil$.

5 Admissible filtrations and exact identification of values

5.1 Observation information and the driving Brownian motion

For deterministic initial conditions the innovation satisfies

$$\widehat{W}_t = \frac{1}{\sigma}\left[Z_t - z_0 - \int_0^t (\widehat{\mu}(\Pi_s) - \sigma^2/2)\,\mathrm{d}s\right]. \tag{5.1}$$

The posterior is adapted to the observation filtration, so this gives $\mathcal{F}^{\widehat{W}}_t \subseteq \mathcal{F}^X_t$. Strong well-posedness gives the reverse inclusion: (Z, Π) is adapted to the completed natural filtration of $\widehat{W}$. Since $Z = \log X$, the completed natural filtrations of X, (Z, Π), and $\widehat{W}$ coincide.

5.2 Why larger admissible filtrations give the same value

Let $\mathcal{H}_t$ be any filtration admitted by the book, in which w remains Brownian and the strong solution is adapted. Its future Brownian increments are independent of $\mathcal{H}_t$, and the future solution is determined by its current state and those increments. Thus the state retains the same Markov transition law relative to $\mathcal{H}$, not merely relative to its natural filtration.

Here is an explicit stopping argument. Put $\widetilde{x}(t) = x(t \wedge \tau_C)$ and $F = -g$, a bounded continuous terminal reward. The stopped time-augmented process is Markov under either filtration and has the same transition operators P_t^{stop}. For a deterministic grid $0 = t_0 < \cdots < t_m = T$, its optimal terminal-reward values satisfy

$$u_m(y) = F(y), \qquad u_j(y) = \max\{F(y), P^{\mathrm{stop}}_{t_{j+1}-t_j} u_{j+1}(y)\}.$$

Backward induction gives the same value under either filtration; a maximizing choice depends only on the current state. This also shows that independent randomization cannot improve the value. Absorption in this argument is only a tool for comparing stopping rules; it is not the diffusion to which local consistency or Theorem 1.1 is applied.

For any $\mathcal{H}$-stopping time $\rho \le \tau_C$, let ρ_m^{grid} be its upward rounding to a deterministic time grid whose mesh tends to zero, and set $\rho_m = \rho_m^{\mathrm{grid}} \wedge \tau_C$. The rounded time is a stopping time for the grid filtration, and $\rho_m \to \rho$ almost surely. Continuity of F and the stopped paths, together with boundedness, gives

$$\mathbb{E}[F(x(\rho_m))] \longrightarrow \mathbb{E}[F(x(\rho))].$$

Each expected reward on the left is bounded above by the natural-filtration continuous stopping value, because the grid values coincide. The same bound therefore holds for every enlarged-filtration stopping rule ρ. The reverse inequality follows by inclusion of the natural stopping rules. This proves equality of the book's weak-formulation value and the observation-filtration value (2.5), also when B_2, B_3 are adjoined.

For the discrete chain (2.10), the filter is a deterministic function of the observed returns, and the returns are recovered from the log-price increments $\Delta Z = \pm\sigma\sqrt{h}$. Its natural state filtration therefore equals the return filtration used in (2.11).

5.3 Value identities and the conclusion of Step 3

All costs, exit restrictions, and admissible values are now identified. Equations (3.5)–(3.6) and the filtration argument give

$$V^h(0, z_0, \pi_0) = -v_R^{(N),M}(x_0, \pi_0), \qquad V(0, z_0, \pi_0) = -V_R^M(x_0, \pi_0). \tag{5.2}$$

Every hypothesis of Theorem 1.1 has been verified. Its conclusion is therefore

$$v_R^{(N),M}(x_0, \pi_0) \longrightarrow V_R^M(x_0, \pi_0) \qquad (N \to \infty). \tag{5.3}$$

This is the precise conclusion used in Step 3 of the main paper's Theorem 4. The estimates in its Steps 1–2 then remove localization and reward truncation, in the order $N \to \infty$, $R \to \infty$, and $M \to \infty$.

The theorem does more than apply ordinary weak convergence to a payoff. Its proof uses joint limits of the chain, auxiliary driving noise, and stopping rules (Chapter 10, Theorem 4.1, pp. 290–291) and comparison with nearly optimal limiting stopping rules (pp. 298–299). These facts control the optimization over stopping times, which convergence of state laws alone would not justify.

6 The randomized filter grid used in Step 4

Let

$$\Gamma_N = \{l/L_N : 0 \le l \le L_N\}, \qquad L_N/\sqrt{N} \longrightarrow \infty.$$

For q between consecutive grid points a, b, let $\mathcal{R}_N(q)$ equal b with probability $(q-a)/(b-a)$ and a otherwise, using a fresh independent auxiliary draw at each step. Then

$$\mathbb{E}[\mathcal{R}_N(q)] = q, \qquad \mathrm{Var}(\mathcal{R}_N(q)) = (q-a)(b-q) \le \frac{1}{4L_N^2}, \qquad |\mathcal{R}_N(q) - q| \le L_N^{-1}.$$

The grid chain uses the same branch probabilities at its current filter value $p \in \Gamma_N$, but replaces $\Phi_\ell^h(p)$ by $\mathcal{R}_N(\Phi_\ell^h(p))$.

6.1 Local consistency and all diffusion-side hypotheses

Conditional on a branch ℓ, the log-price increment is fixed and rounding has mean zero relative to the unrounded posterior. Thus the means of $\Delta Z, \Delta\Pi$ and their cross moment are unchanged at each common current state. The difference between the rounded and unrounded filter second moments is exactly

$$\sum_{\ell \in \{u,d\}} p_\ell^h(p)\, \mathrm{Var}(\mathcal{R}_N(\Phi_\ell^h(p))) \le \frac{1}{4L_N^2}.$$

After division by $h = T/N$, this is at most

$$\frac{1}{4hL_N^2} = \frac{1}{4T}\left(\frac{\sqrt{N}}{L_N}\right)^2 \longrightarrow 0.$$

The jump bound becomes $h + K\sqrt{h} + L_N^{-1} \to 0$. These are the estimates in the paper's Lemma 4. The transition probabilities and entire price law are not asserted to be unchanged along different chains; the comparison is at a common current state, and all estimates are uniform.

There are at most four successors per grid-chain state, so the reachable time-augmented states before exit are again finite. The limiting coefficients, costs, uniqueness, exit continuity, and exit-moment checks are the same as above. The grid chain's natural filtration includes its realized rounded states, as required by the Bellman algorithm. Its price-moment and localization bounds use the same uniform conditional increment bounds as the exact chain.

6.2 Initial interpolation and varying initial states

The fixed initial state required by Theorem 1.1 can be retained even if grid starting points vary. For any $\pi_N \in \Gamma_N$ with $\pi_N \to \pi_0$, put $d_N = \pi_0 - \pi_N$ and translate the chain by

$$(nh, Z_n, \Pi_n^{\Gamma_N}) \longmapsto (nh, Z_n, \Pi_n^{\Gamma_N} + d_N).$$

It now starts at the fixed point $(0, z_0, \pi_0)$. Its increments are unchanged. The original local moment limits are evaluated at the unshifted posterior; global Lipschitz continuity changes them by $O(|d_N|)$ after scaling by h, which tends to zero. The posterior stays in $[d_N, 1+d_N] \subset (-1, 2)$ for all sufficiently large N. Thus translation changes neither the obligatory exit index, the reward, nor the natural filtration up to a deterministic invertible relabeling. Theorem 1.1 applies and gives the same bounded, localized limiting value.

The algorithm's reported initial value is the convex combination of the values at the two grid points bracketing π_0. The initial randomized grid point is observed by the stopping agent; the reported number is its expectation before that rounding is observed. Both bracketing points tend to π_0, so both localized values converge to $V_R^M(x_0, \pi_0)$, and their convex combination does too. The uniform moment and localization estimates then give the grid-value assertion of the paper's Theorem 4.

References

[1] H. J. Kushner and P. Dupuis, *Numerical Methods for Stochastic Control Problems in Continuous Time*, 2nd ed., Applications of Mathematics 24, Springer, New York, 2001.

[2] I. Karatzas and S. E. Shreve, *Brownian Motion and Stochastic Calculus*, 2nd ed., Graduate Texts in Mathematics 113, Springer, New York, 1991.

[3] V. Golomoziy, K. Kladívko, and Y. Mishura, *Binomial Trees for Optimal Stopping Problems under Drift Switching and Partial Information*, main paper, working manuscript. The present note verifies the application in Theorem 4, Step 3, and the grid extension in Step 4. Supplementary-material repository: `https://github.com/vitaliy-golomoziy/non-recombining-binomial-tree`.